BUSINESS MEMORANDUM

TO:	Product & Retention Leadership	**DATE:**	June 2026
FROM:	Anukrati Rajawat — MS Business Analytics, UNCW	**MODEL:**	Random Forest (Acc: 78% | AUC: 0.809)
SUBJECT:	**Customer Churn Analysis & Data-Driven Retention Strategy**		

Executive Summary & At-a-Glance Results

This memorandum outlines a comprehensive customer churn analysis conducted using the IBM Telco Customer Churn dataset. By deploying an optimized machine learning framework, we identified key operational and behavioral drivers of attrition. More importantly, this analysis shifts the retention strategy from a reactive framework to a proactive, high-ROI business intervention model.

Operational Metric	Value / Impact
Total Customers Analyzed	7,032 customers (21 features)
Overall Baseline Churn Rate	**26.6% (1,869 customers)**
Total Revenue at Direct Risk	$2,862,927 (17.8% of total revenue)
High-Risk Customers Flagged by Model	1,537 (21.9% of customer base)
Projected Model Revenue Saved	**$1,800,872**
Retention Outreach Budget / Cost	$46,450
Net Business Value Generated	**$1,754,422**

Key Findings & Diagnostics

Finding 1: Contract Type is the Primary Driver of Attrition

Month-to-month contract holders exhibit a severe churn rate of 42.7%, which is 15 times higher than long-term 2-year contract holders (2.8%). One-year contract customers sit at a manageable 11.3%. Crucially, 55% of our entire customer base (3,875 customers) are currently on month-to-month terms, making this segment our highest-priority operational vulnerability.

Finding 2: Onboarding Bottlenecks in Early Tenure Cohorts

Customers in their first 12 months present a significantly elevated risk profile compared to long-tenured accounts. This indicates an immediate onboarding and early-stage engagement gap: customers are terminating service before fully realizing the ecosystem's value. Retention investments during Year 1 represent our highest potential ROI.

Finding 3: Financial Sensitivity Among High Monthly Premium Accounts

Accounts with monthly charges exceeding $65 who are concurrently on month-to-month contracts constitute the most volatile customer cluster. Lacking structural barriers to exit, these premium customers respond directly to financial pain points or competitive market promotions, resulting in immediate churn.

Finding 4: Substantial Financial Exposure ($2.86M Revenue at Risk)

The total historical charges accumulated by churned customers amount to $2,862,927 (17.8% of total corporate revenue). By tuning our predictive model to catch 62% of impending churners, proactive account interventions are mathematically positioned to reclaim up to $1.8M in revenue.

Predictive Model Performance & Optimization

A Random Forest classifier was trained on 5,625 records and validated against an independent holdout set of 1,407 records. TotalCharges was intentionally excluded from the feature space to eliminate data leakage risks, as it serves as a direct linear mathematical derivative of MonthlyCharges multiplied by tenure.

Performance Metric	Default Threshold (0.50)	Optimized Threshold (0.35)
Overall Accuracy	78.2%	74.6%
Churn Recall (Sensitivity)	50.8%	**66.3% (+15.5%)**
Churn Precision	60.7%	51.7%
ROC-AUC Score	0.809	0.809

Strategic Threshold Tuning Note: The operational decision threshold was lowered from 0.50 to 0.35. In subscription business mechanics, a False Negative (failing to identify an account that will churn) carries significantly heavier financial consequences than a False Positive (allocating a minor retention concession to a stable account). This adjustment captures an extra 230 high-risk accounts.

Strategic Recommendations

Recommendation 1: Contract Transition Incentive Campaign

Target the 3,875 month-to-month contract holders precisely at their 6-month operational mark. Deploy an automated upgrade campaign offering a 1- or 2-year contract coupled with a 10-15% monthly discount or a high-value complimentary digital service add-on. Achieving a modest 20% conversion rate will successfully suppress baseline churn by ~6 percentage points.

Recommendation 2: Structural First-Year Lifecycle Management

Target newly acquired accounts to institutionalize an automated onboarding journey covering months 1–12. Mandate key programmatic customer check-ins: an executive welcome concierge call at Day 7, a digital operational pulse-check at Month 3, and a comprehensive value-realization review at Month 6. Proactive engagement in this critical window protects lifetime value.

Recommendation 3: Predictive Tactical Outreach for the Current Quarter

The analytics pipeline has isolated 1,537 accounts with critical churn scores (>66%). Operationalize the customer success team to prioritize outreach to the top 20% tier containing the highest combination of risk and monthly spend. This campaign provides an estimated 38x marketing ROI ($46,450 deployment cost vs. $1,800,872 saved revenue).

Technical Stack & Reproducibility:
Predictive pipeline developed in Python utilizing scikit-learn and the Random Forest architecture. Interactive executive dashboards engineered in Power BI for operational deployment.
Code Repository & Documentation: https://github.com/AnukratiR/customer-churn-prediction